Aryan Kumar

B.Tech Computer Science Engineering (CSE)
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Summary

Computer Science Engineering undergraduate with a strong focus on **AI systems, numerical simulation, and energy-aware computation**. Interested in how **physical modeling, algorithmic design, and system-level constraints** shape performance, stability, and scalability. Actively building **simulation-driven and research-oriented projects** emphasizing correctness, interpretability, and computational efficiency.

Education

B.Tech — Computer Science Engineering *May 2025 – May 2029*
Dronacharya College of Engineering, Delhi

Technical Skills

Languages: C, Python
Core Areas: Numerical Simulation, AI Systems, Energy-Aware Computing, Algorithms
Technical Concepts: Newtonian Mechanics, Ordinary Differential Equations (ODEs), Runge–Kutta Methods, Time-Stepping Schemes, Model Scaling Laws

Projects

N-Body Gravitational Simulation (C) — GitHub

- Developed a Newtonian N-body simulator modeling pairwise gravitational interactions in three-dimensional space.
- Implemented softened inverse-square force computation to prevent numerical singularities at small separations.
- Advanced system dynamics using semi-implicit Euler integration with configurable timestep and body count.
- Evaluated numerical stability and physically consistent behavior over extended simulation horizons.

Double Pendulum Simulation (C + Python) — GitHub

- Built a numerical simulation of a nonlinear double pendulum system exhibiting deterministic chaos.
- Implemented fourth-order Runge–Kutta (RK4) integration for accurate time evolution of coupled equations.
- Designed a clean separation between physics computation (C) and visualization and analysis (Python).
- Analyzed sensitivity to initial conditions via trajectory divergence and phase-space behavior.

Hierarchical Multi-Agent Framework for Sustainable AI — GitHub

- Proposed a three-tier hierarchical multi-agent architecture for energy-efficient AI-assisted software development.
- Defined explicit roles for orchestration, domain-specialized execution, and guardian-based oversight.
- Modeled per-task energy consumption as a function of model power draw and execution time.
- Framed model selection as an optimization problem under quality and latency constraints using structured communication.

Additional

Interests: GPU architectures, AI acceleration, numerical physics, sustainable computing
Languages: English, Hindi
Status: Open to internships and research-oriented roles